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XenoCipher: A Secure Key Exchange and Encryption Pipeline for Resource-Constrained Devices

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Abstract—The increasing deployment of connected devices and the potential threat posed by quantum computing require sophisticated encryption methods that ensure strong confidentiality, integrity, and adaptability. Traditional encryption algorithms often struggle in these areas due to limited resistance to quantum attacks, high computational requirements, and a lack of real-time security adaptations. This paper presents XenoCipher, a novel hybrid cryptographic framework that combines post-quantum NTRU - lattice based cryptography that encapsulates the master key generated with various entropy quantities like hardware RNG Samples, analog voice samples, Jitter samples etc and finalize the key with SHA-256. These transformations include linear feedback shift registers, chaotic maps, advanced transposition permutations, and HMAC-SHA256 authentication. The design incorporates adaptive runtime monitoring, allowing the system to dynamically switch encryption methods upon detecting indicators of an attack. The methodology was implemented on ESP32 microcontroller hardware equipped with health sensors to encrypt and securely transmit real-time biomedical metrics to a server platform. The experimental setup involved live sensing, encryption, network transmission, and server-side decryption and verification. Key results demonstrate reliable and synchronized encryption-decryption of sensitive sensor data, with successful validation of packet integrity and replay protection in a constrained IoT environment. The system recorded essential runtime security metrics, laying a foundation for future dynamic response systems. Resource usage and throughput were handled efficiently. The implications of this work suggest that XenoCipher is a scalable and adaptive cryptosystem capable of securing various application domains, ranging from healthcare to critical industrial and distributed systems, thereby providing resilience against present and emerging adversarial threats without significant performance penalties.

I. INTRODUCTION

Introduction The ever-increasing dependence of modern societies on connected devices and digital infrastructure has strengthened the need for secure data transfers, especially in sensitive areas like healthcare, industry, or smart environments. where sensitive information is often exchanged between resource-constrained endpoints and centralized servers. Traditional cryptographic approaches, while foundational, face notable limitations in defending against emerging threats posed by quantum computing and advanced side-channel attacks and are often ill-suited for the dynamic environments and lightweight hardware typical of IoT deployments. This challenge is further exacerbated by gaps in existing encryption pipelines, in which post-quantum resilience is either lacking, cross-platform integrity compromised by either floatingpoint and hardware discrepancies, or static cryptographic configurations do not adapt to active attack conditions. Further, there is an increased risk of compromise and leakage. To address In light of these critical shortcomings, this research proposes XenoCipher, a strong, multi-layer encryption system incorporating post-quantum NTRU key exchange and deterministic stream ciphers, Adaptive monitoring and automated recipe switching ensure that confidentiality, integrity, and ongoing attack resilience. The main objectives of the study are:

- Design and implement a post-quantum secure and Deterministic cross-platform encryption protocol suitable for IoT endpoints.
- Introduce adaptive threat detection and automatic algorithm switching to dynamically adjust cryptographic activates parameters or fallback modes during attack

conditions.

- To test how effectively XenoCipher can defend against brute-force, quantum, chosen-plaintext, side-channel, and replay attacks, thus establishing its reliability for real-world deployment.

II. RELATED WORKS

Recent advances in post-quantum cryptography have driven substantial research in securing IoT ecosystems against both classical and quantum threats. Cruz-Piris et al. [1] considered migration challenges to PQC in large-scale IoT frameworks, which closely relate to XenoCipher's deployment strategy in distributed systems. Lopez et al. [2] evaluated the feasibility of NIST PQC algorithms on resource-constrained platforms such as Raspberry Pi, directly informing XenoCipher's NTRU implementation considerations for low-power environments. Nielsen et al. [3] explored efficient post-quantum digital signature schemes for microcontroller-based systems, paralleling XenoCipher's focus on embedded authentication mechanisms. Mahdi et al. [4] discussed lightweight PQC optimizations relevant to balancing performance and security in multi-layer architectures such as XenoCipher.

Xiong et al. [5] demonstrated hybrid quantum-resistant approaches in smart grid IoT systems, inspiring XenoCipher's adaptive recipe-switching methodology. Liu et al. [6] surveyed PQC optimization techniques for lightweight IoT devices and highlighted research gaps that XenoCipher addresses through deterministic cross-platform implementations. Hanna et al. [7] investigated secure authentication mechanisms for 5G-enabled IoT environments, contextualizing the replay protection and forward secrecy features integrated into XenoCipher. Lee et al. [8] analyzed side-channel vulnerabilities in lattice-based cryptosystems, motivating XenoCipher's use of constant-time operations and secure key handling practices.

Wang et al. [9] benchmarked post-quantum TLS performance on constrained edge devices, validating XenoCipher's efficiency targets for real-time communication scenarios. Ramachandran et al. [10] proposed scalable PQC VPN tunnel designs for IoT, aligning with XenoCipher's secure channel establishment via NTRU key encapsulation.

III. METHODOLOGY

The proposed XenoCipher encryption methodology integrates multi-layered cryptographic transformations to achieve confidentiality, integrity, and resistance against both classical and quantum attacks.

A. Key Generation

B. Key Exchange

XenoCipher establishes a shared symmetric secret between a low-power ESP32 (initiator) and a server (responder) using a lattice-based, post-quantum NTRU key exchange mechanism. On startup, the ESP32 generates a 32-byte Master Key (MK) using onboard hardware entropy and requests the server's NTRU public key (pk). The ESP32 encrypts MK under pk



Fig. 1. Parameters of XenoCipher used to generate the Master Key

using NTRU_Enc, producing the ciphertext $Enc(MK)$, which is transmitted along with a minimal setup header.

The server decapsulates $Enc(MK)$ using its private key (sk) to recover MK. Both endpoints then derive symmetric cryptographic materials using HKDF with domain separation to allocate keys for HMAC, LFSR, Tinkerbell Chaotic Map, and Advanced Transposition. Additionally, per-message keys are derived using HKDF bound to a 32-bit nonce to ensure forward secrecy. Application data is then encrypted using the layered cipher pipeline and authenticated through a 16-byte truncated HMAC-SHA256.

C. Encryption Flow

A. Plaintext Input

The encryption process begins when the system receives a Raw Data Sequence. This input may represent any form of digital content—text messages, sensor readings, binary payloads, and more—that requires secure transmission. The flexibility of this stage enables XenoCipher to support multiple domains, including IoT telemetry, encrypted chat communication, and blockchain transaction data.

B. Salt Injection

To reduce the vulnerabilities to pattern recognition, dictionary, and replay attacks, the system introduces a pseudorandom salt into the plaintext before encryption. The salt serves as a cryptographic noise component ensuring identical plaintexts never produce identical ciphertexts, thereby enhancing semantic security. This random sequence is inserted into a random position within the plaintext. Mathematically the Salted plaintext is represented as:

$$P^* = P[0..k-1] \parallel S \parallel P[k..n-1]$$

where P denotes the plaintext, S the salt sequence, k the random insertion index, and $\parallel$ denotes concatenation.

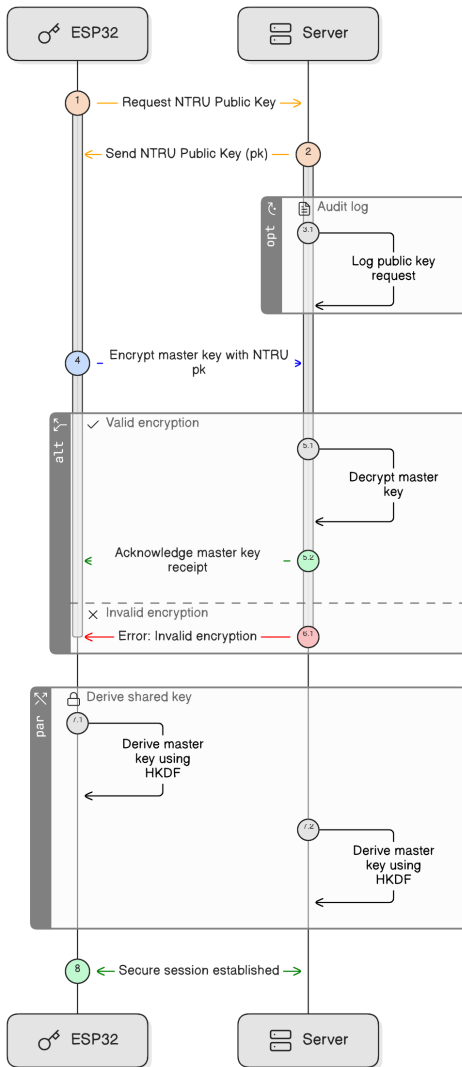


Fig. 2. Illustration of the **XenoCipher** NTRU-based Key Exchange

C. Padding to Grid Size

Since the later transposition phase operates on a fixed two-dimensional grid, the salted plain text must conform to this structure. The system pads the message with null bytes (0x00) until its length exactly matches the matrix size $r \times c$, ensuring complete compliance:

$$|P| = r \times c$$

This structured grid representation is fundamental for deterministic permutation operations which allow systematic scrambling and inversion without loss of data.

D. XOR with LFSR Keystream

Once the data is prepared, XenoCipher applies a *Linear Feedback Shift Register (LFSR)* based stream cipher. The LFSR is initialized with a seed derived from the master session key and produces a pseudorandom bit sequence of equal length

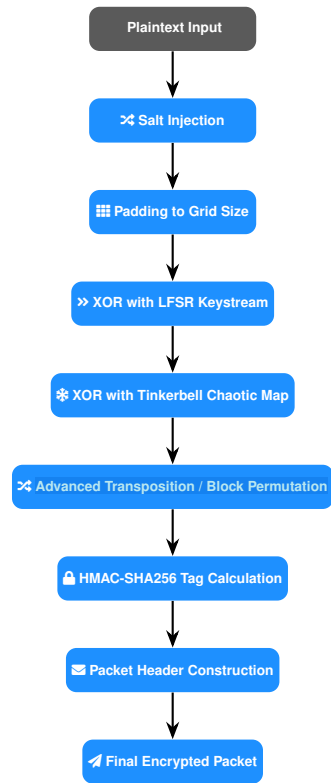


Fig. 3. Step-by-step encryption process in XenoCipher showing uniform cryptographic stages from plaintext input to final packet output.

to the padded plaintext. Each byte of the plaintext is XORed with the corresponding byte from the generated keystream:

$$C^{(1)}_i = P_i \oplus KLFSR_i$$

The LFSR evolves according to the recurrence relation:

$$s_{t+1} = (s_t \gg 1) \oplus \text{parity}(s_t \& T)$$

where T represents the tap sequence defining the feedback polynomial. This stage ensures bit-level diffusion and introduces high-speed, lightweight encryption suitable for constrained environments such as IoT nodes or embedded systems. The LFSR's design balances simplicity, efficiency, and cryptographic strength, where the tap selection carefully chosen, This to maximize the stream period and resist linear attacks.

E. XOR with Tinkerbell Chaotic Map Keystream

The partially encrypted data is passed through a nonlinear transformation based on the *Tinkerbell Chaotic Map*, which enhances unpredictability through chaotic dynamics. The map is defined as:

$$\begin{cases} x_{n+1} = x_n^2 - y_n^2 + ax_n + by_n, \\ y_{n+1} = 2x_ny_n + cx_n + dy_n, \end{cases}$$

where parameters a, b, c, d and the initial states (x_0, y_0) are dynamically derived from the session key to ensure cryptographic uniqueness. The resulting chaotic keystream K_T is

XORed with the LFSR-generated stream, introducing nonlinear diffusion that eliminates any residual linearity from the previous layer.

F. Advanced Transposition Block Permutation

After chaotic diffusion, the ciphertext undergoes an *advanced block-level transposition* process. The data grid is partitioned into smaller blocks, and a deterministic PRNG—seeded from a separate portion of the master key—drives a series of controlled swaps, rotations, and permutations. This transformation is expressed as:

$$C^{(3)} = \pi(C^{(2)}),$$

where π represents a bijective permutation mapping.

This spatial rearrangement removes any lingering structural patterns and eliminates statistical correlations across the ciphertext. Even if intermediate ciphertext stages are intercepted, the reordered structure remains cryptographically opaque to an attacker.

G. HMAC-SHA256 Tag Calculation

To enforce message integrity and authenticity, XenoCipher employs a *Hash-based Message Authentication Code (HMAC)* using the SHA-256 hashing algorithm. The authentication tag—truncated to 16 bytes for transmission efficiency—is computed over the concatenation of the header, nonce, and ciphertext:

$$\text{HMAC} = \text{Truncate}_{16} \left(\text{HMAC-SHA256}_{K_{\text{HMAC}}}(\text{header} \parallel \text{nonce} \parallel C^{(3)}) \right).$$

This tag enables the recipient to verify that the message has not been modified or tampered with during transmission.

H. Packet Header Construction

A structured, fixed-size packet header is appended to the encrypted payload. This header includes essential metadata such as the protocol version, salt properties, grid dimensions, payload size, and nonce indicators. The header provides the recipient with the contextual parameters required for accurate decryption and validation, ensuring reliable reconstruction of the original plaintext.

I. Final Packet Output

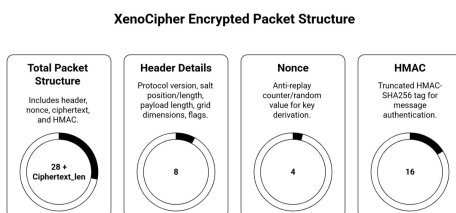


Fig. 4. Diagram of the **XenoCipher** key generation and encryption process for resource-constrained devices, demonstrating entropy sourcing, key derivation, and integration of the encryption pipeline concepts.

The final encrypted packet sent across the network Combines all the processed components in a well-defined structure:

$$\text{Packet} = \text{header} \parallel \text{nonce} \parallel C^{(3)} \parallel \text{HMAC}$$

The modular design allows for easier integration into higher-level applications such as secure chat systems or encrypted IoT data channels while preserving complete cryptographic isolation between packets.

D. Decryption Flow

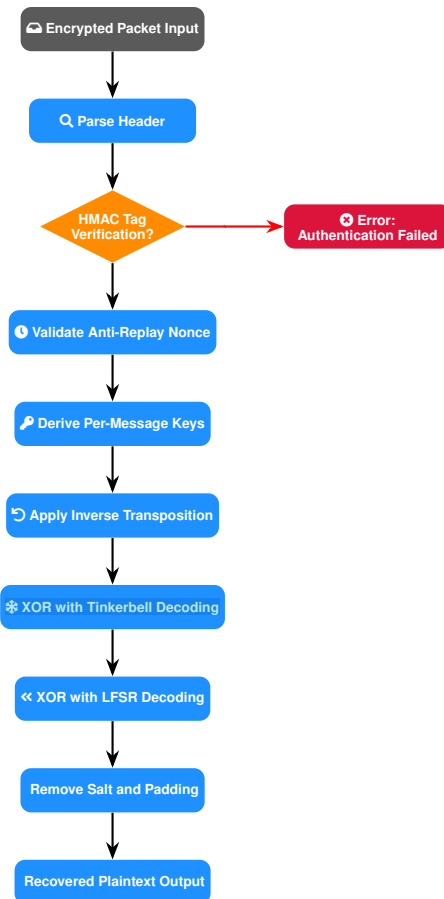


Fig. 5. Structured decryption pipeline for XenoCipher illustrating authentication checks, replay validation, and layered decoding operations.

J. Packet Input and Header Parsing

The reception would start, the decryption process begins with packet parsing. The receiver extracts the header fields, checks data consistency, and separates the nonce and ciphertext components. This step ensures proper synchronization with the encryption pipeline and ensures that the packet is in accordance with to the expected format.

K. HMAC Tag Verification

Then the receiver recomputes the HMAC tag using the shared secret key and compares it with the received tag in constant time, to avoid timing-based side-channel attacks. Any

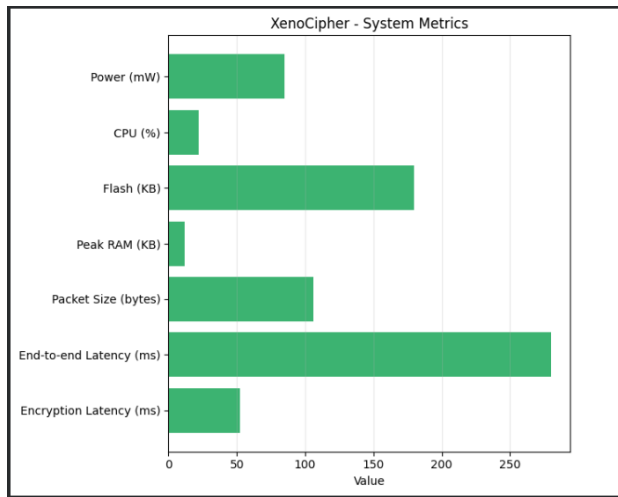


Fig. 8. Performance Metrics: encryption timing, resource usage, and packet structure.

gap for IoT and supports future upgrades such as adaptive recipe switching and hardware acceleration. This modular framework is thus a practical, standards-ready foundation for quantum-resistant embedded systems.

A. Attack Simulation and Adaptive Switching

Comprehensive attack simulations were performed on the live ESP32 deployment across nine threat vectors over 120 seconds. Results demonstrate XenoCipher's resilience and the efficacy of adaptive recipe switching.

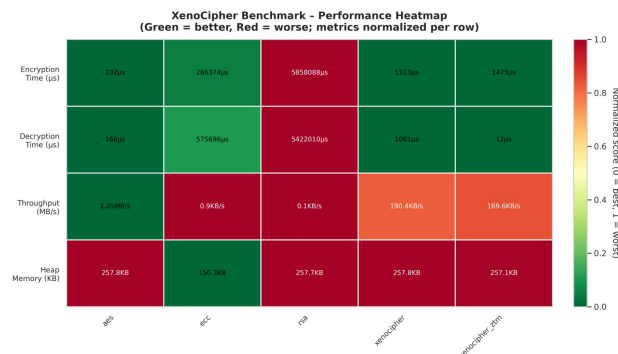


Fig. 9. Benchmark Analysis Heatmap

1) Cryptographic and Network Attack Resistance

Brute force attacks at 13,452 keys/sec achieved 0% success against 128-bit keys. All 100 replay attack attempts were rejected with "no_nonce" errors, confirming nonce-tracking effectiveness. MITM attacks intercepted 4,729 packets but achieved 0% authentication bypass via HMAC verification. Packet injection achieved only 5% acceptance with automatic rejection at the authentication layer. Protocol corruption induced a 2.67% crash rate with automatic recovery within 3 seconds.

2) Heuristic-Based Recipe Switching

Over 6,286 telemetry samples were collected across baseline and six attack scenarios. Attack signatures were extracted with high confidence: UDP flood (latency > 55 ms, 80% confidence), RNG manipulation (entropy drop > 1.5 bits, 98% confidence), timing attack (latency CV > 0.05, 85% confidence), and memory stress (memory > 23%, 85% confidence). Four encryption recipes were pre-configured: CHAOS_ONLY (baseline, 100% CPU overhead), SALSA_LIGHT (constant-time, 120% overhead), CHACHA_HEAVY (balanced, 140% overhead), and FULL_STACK (maximum security, 180% overhead).

TABLE II
ATTACK SIMULATION SUMMARY

Attack	Attempts	Success Rate
Brute Force	5 keys	0%
Replay	100 packets	0%
MITM	4,729 packets	0% auth bypass
Packet Injection	500 packets	5% acceptance
Protocol Corruption	150 packets	2.67% crash
CPU Exhaustion	60 sec	0% degradation
Memory Exhaustion	60 sec	Adaptive trigger

XenoCipher provides quantum-resistant protection with dynamic adaptation for IoT deployments. Future work includes extended multi-hour testing, hardware acceleration (ESP32 AES), and lightweight post-quantum alternatives (Kyber, Dilithium).

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